Mrugank Dake

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Hanover, NH

ACADEMIC HISTORY

Dartmouth College — Hanover, NH
Postdoctoral Researcher in Psychological and Brain Sciences — **June 2026 – Present**
Advisor: Dr. John Murray

New York University — New York, NY
PhD in Computational Cognitive Neuroscience — **May 2026**
Advisor: Dr. Clayton Curtis
Dissertation: The nature and role of distributed representation in Working Memory

MPhil in Experimental Psychology — **May 2025**

Indian Institute of Science Education & Research (IISER) — Tirupati, India
BS-MS Dual Degree (Biology & Physics) — **August 2021**
Advisor: Dr. Nandini Rajamani
Master's Thesis: Statistical and Machine learning approaches to Squirrel Acoustics

SPECIALIZED TRAINING

- **Radical Ventures** — Virtual
  Radical AI Founders Masterclass — **2026**
- **Marine Biological Laboratory (MBL)** — Woods Hole, MA
  Methods in Computational Neuroscience (MCN) Course — **2026**
- **Neuromatch Academy** — Virtual
  Deep Learning Course — **2022**

PUBLICATIONS

- **Dake M.**, Dandekar S., Curtis C. E. (2026). Neural synchrony between prefrontal and visual cortex supports visual working memory. *bioRxiv*
- **Dake M.**, Curtis C. E. (2025). Perturbing human V1 degrades the fidelity of visual working memory. *Nature Communications*.

SOFTWARE & TOOLS

- **pRF fitting toolbox** — JAX-based toolbox for population receptive field (pRF) modeling of fMRI data using the compressive spatial summation (CSS) model; fits a full volumetric dataset in ∼5 min on a single GPU. *In development.*
- **Proxenos** — Open-source, privacy-first agent that keeps a person's own records on their computer and drafts the next step (emails, forms, follow-ups) for them to send; built first for people getting through the healthcare system. Desktop app for Mac and Windows. mindemory.io/proxenos
- **Thanos** — A lab's memory: what a research group works out over the years, from grant and account details to equipment and where the code lives, kept in one place to search after people move on. Started for the Murray Lab (Dartmouth) and the Clay Space Lab (NYU). *In development.* mindemory.io/thanos
- **MCN Network Graph** — Interactive 2D and 3D visualization of the research network of the MBL Methods in Computational Neuroscience cohorts. mindemory.io/projects/mcn-network
- **Alexαndria** — Platform integrating shared bibliographies with a collaborative activity space for research teams. *Prototype.* alexandriasocial.net
- **Notion Paperpile Integration** — Turns a Paperpile library into a Notion reading list, and maps it by topic and co-authorship in 2D and 3D. github.com/mindemory/notion_paperpile
- **EEG/MEG Methods course materials** — Open tutorials and analysis notebooks for the NYU iEEG/EEG/MEG Methods PhD course (2024). github.com/mindemory/EEG_MEG_Methods

TALKS

- **Dake M.**, Dandekar S., Curtis C. E. Neural synchrony between prefrontal and visual cortices supports visual working memory; *Working Memory Symposium (Virtual)* **July 2026**
- **Dake M.**, Curtis C. E. A tale of two cortices; *NYU TELL GSAS competition* **April 2025**
- **Dake M.**, Curtis C. E. Perturbing human V1 degrades the fidelity of visual working memory; *Working Memory Symposium (Virtual)* **July 2024**

POSTERS

- **Dake M.**, Dandekar S., Curtis C. E. Working memory synchronizes oscillations within and across cortical areas; *Society for Neuroscience (SfN), San Diego CA, USA* **Nov 2025**
- **Dake M.**, Dandekar S., Curtis C. E. Working memory synchronizes oscillations in visual cortex; *Computational Cognitive Neuroscience (CCN), Amsterdam, Netherlands* **May 2025**
- **Dake M.**, Curtis C. E. Do perturbations to visual cortex impact working memory?; *Society for Neuroscience (SfN), Washington D.C., USA* **Nov 2023**
- Khare S., Chopra S., Grover H., Vohra T., Mahmood U., Jatwani K., Kaushik D., Pandey A., Bhardwaj S., Tanwar A., Singh J., Shekhar C., **Dake M.**, Krishnamoorthy K., Kumar K. Li-Koff: To Detect and Degrade N-Nitrosamines; *Giant Jamboree (iGEM), Boston, USA* **Nov 2020**
- Amar I., Avadhani K., Bajaj M., Balasubramanian D., Bhagat S., Chutani N., **Dake M.**, Khatri U., Krishna N., Jacob M., Mal S., Mohapatra O., Pal A., Saha D., Tripathy B., Mukherjee R., Rao B.J. CoCa coli: probiotic immunotherapy against colon cancer; *Giant Jamboree (iGEM), Boston, USA* **Nov 2019**

PRE-DOCTORAL RESEARCH EXPERIENCE

- Statistical and Machine learning approaches to Squirrel Acoustics (Master's Thesis) **Aug 2020 - Aug 2021**
  Advisor: Dr. Nandini Rajamani, IISER Tirupati, India
- EEG and MRS Analysis for underpinning neural correlates of attentional blink **June - Aug 2019**
  Advisor: Dr. Sridharan Devarajan, IISc Bangalore, India
- CoCa coli: probiotic immunotherapy against colon cancer **Jan 2018 - Nov 2019**
  Advisors: Dr. Raju Mukherjee; Dr B J Rao, IISER Tirupati, India
- Isolation and characterization of neurons and glia from goat hippocampal culture **Sep 2018 - May 2019**
  Advisor: Dr. Vasudharani Devanathan, IISER Tirupati, India
- Identifying differentially expressing tumor suppressor genes in *Drosophila melanogaster* wing and haltere imaginal discs **May - July 2018**
  Advisor: Prof. L S Shashidhara, IISER Pune, India
- RNA Origami (Square) using in-vitro replication and transcription **Jan - March 2018**
  Advisor: Dr. Ashwani Sharma, IISER Tirupati, India
- Head direction cell model for rodents with internal updating mechanism **May - July 2017**
  Advisor: Dr. Collins Assisi, IISER Pune, India

PROFESSIONAL & TEACHING EXPERIENCE

Meta Reality Labs Burlingame, CA
Product Data Science (PhD) Intern **May – Aug 2025**

New York University New York, NY
Teaching Assistant: Scientific Programming (MA Course) **Fall 2025**
Teaching Assistant: iEEG/EEG/MEG Methods (PhD Course) **Fall 2024**
Teaching Assistant: Introduction to Cognitive Neuroscience (UG) **Fall 2023**
Teaching Assistant: Math Tools III: Linear Systems (PhD Course) **Spring 2023**

PROFESSIONAL SERVICE

Ad-hoc Reviewer: Brain Structure & Function (2026), eLife, ICLR (2026), Cognitive Science Society (2025), Computational Cognitive Neuroscience (CCN 2025, 2026)
Website Design Coordinator: Climatematch Academy **Aug – Dec 2023**
Project Mentor: iGEM, Manav Rachna International Institute of Research and Studies **2020**

Student Team Leader: iGEM, IISER Tirupati **2019**

AWARDS

Society for Neuroscience Trainee Professional Development Award ($1,000)	**2025**
Dean's Award for Research Communication Excellence, NYU ($2,000)	**2025**
GSAS Conference Travel Grant, NYU	**2025**
MacCracken Doctoral Fellowship, NYU	**2021 – 2026**
Indian Biological Engineering Competition (iBEC) Grant (∼$12,000 each)	**2019 – 2020**
INSPIRE Fellowship, Department of Science and Technology India	**2016 – 2021**
Regional Winner, 5th Place, Indian National Brain Bee (INBB)	**2015**

SKILLS

Programming
Python, MATLAB, R, SQL, LaTeX, CSS, JavaScript, HTML, React, NodeJS, Supabase, Vercel

Neuroscience Tools
EEG, MEG, fMRI, TMS, ECoG, Fieldtrip, MNE-Python, SimNIBS, Psychtoolbox

Data Science & ML
JAX, PyTorch, TensorFlow, Keras, Scikit-learn, Neural Networks, Time-series Analysis

Other Tools & Languages
Git/GitHub, Adobe Illustrator, Adobe Photoshop
English (Fluent), Hindi, Marathi